

8)

a) Prove that the input impedance of a quarter wave line is $Z_{in} = R_0^2 / Z_R$

The general expression for input impedance, is, at high freq

$$Z_{in} = Z_0 \left[\frac{Z_R + j Z_0 \tan(\beta l)}{Z_0 + j Z_R \tan(\beta l)} \right] \rightarrow \textcircled{1}$$

The condition for a quarter-wave line is

$$d = \lambda/4$$

where,

$$\beta = \frac{2\pi}{\lambda}$$

$$\beta l = \frac{2\pi}{\lambda} \times \frac{\lambda}{4}$$

$$\boxed{\beta l = \pi/2}$$

$\textcircled{1} \Rightarrow$

$$Z_{in} = Z_0 \left[\frac{\frac{Z_R}{\tan(\beta l)} + j Z_0}{\frac{Z_0}{\tan(\beta l)} + j Z_R} \right]$$

$$= Z_0 \left[\frac{j Z_0}{j Z_R} \right]$$

$$Z_{in} = \frac{Z_0^2}{Z_R}$$

$$\therefore Z_0 = R_0$$

$$\boxed{Z_{in} = \frac{R_0^2}{Z_R}}$$

$$\tan(\pi/2) \rightarrow \infty$$

Hence we take

$\tan(\pi/2)$ in common

$$\frac{\text{Anything}}{\infty} = 0$$

b) Design a quarter wave transformer to match a load of 200Ω to a source resistance of 500Ω operating freq is 200 MHz .

Given,

$$Z_{in} = Z_S = 500\Omega$$

$$Z_R = 200\Omega$$

$$f = 200\text{ MHz}$$

Design

W.K.T,

$$Z_{in} = \frac{Z_0^2}{Z_R}$$

$$Z_0 = \sqrt{Z_{in} Z_R}$$

$$= \sqrt{500 \times 200}$$

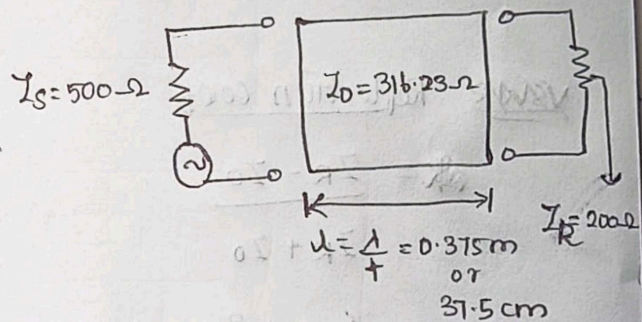
$$\boxed{Z_0 = 316.23\Omega}$$

$$\lambda = \frac{c}{f} = \frac{3 \times 10^8}{200 \times 10^6} = 1.5\text{ m}$$

$$\boxed{\lambda = 1.5\text{ m}}$$

$$l = \frac{\lambda}{4} = \frac{1.5}{4} = 0.375\text{ m}$$

$$\boxed{l = 0.375\text{ m}} \Rightarrow 37.5\text{ cm}$$



To match 200Ω load to the 500Ω source at 200 MHz , we a transmission line section with a characteristic impedance $Z_0 = 316.23\Omega$ and a length of 37.5 cm

10) a) A low loss high frequency transmission line having a characteristic impedance of 500Ω is terminated by an impedance $600 + j300 \Omega$. Determine the VSWR & Reflection co-efficient.

Sln:

Given,

$$Z_0 = 500 \Omega$$

$$Z_R = 600 + j300$$

VSWR Reflection coef

$$K = \frac{Z_R - Z_0}{Z_R + Z_0}$$

$$= \frac{600 + j300 - 500}{600 + j300 + 500}$$

$$= \frac{100 + j300}{1100 + j300}$$

$$K = 0.15 + j0.23$$

Reflection co-eff

VSWR

$$S = \frac{1 + |K|}{1 - |K|}$$

$$|K| = \sqrt{(0.15)^2 + (0.23)^2}$$

$$= \sqrt{0.0754}$$

$$|K| = 0.275$$

$$S = \frac{1 + 0.275}{1 - 0.275}$$

$$S = 1.758$$

b) A lossless line in air having a characteristic impedance of 300Ω is terminated in unknown impedance. The 1st voltage minimum is located at 15 cm from the load. The standing wave ratio is 3.3. Calculate the terminated impedance.

soln:

Given,

$$Z_0 = 300 \Omega$$

$$S = 3.3$$

$$K = \frac{S-1}{S+1}$$

$$= \frac{2.3}{4.3} = 0.534$$

W.K.T,

$$K = \frac{Z_R - Z_0}{Z_R + Z_0}$$

$$Z_R = -Z_0 \left(\frac{K+1}{K-1} \right)$$

$$Z_R = -300 \left(\frac{1.534}{-0.466} \right)$$

=

$$K = \frac{S-1}{S+1} = \frac{2.3}{4.3} = 0.534$$

1st voltage min at 15 cm (For a voltage min to occur at the load, the reflection co-ef must be -ve)

$$K = -0.534$$

$$Z_R = -300 \left(\frac{0.466}{-1.534} \right)$$

$$Z_R = 91.13 \Omega$$

stub length and stub location

$$Z_L = 115 + j15 \Omega$$

$$Z_0 = 50 \Omega$$

$$Z_L = \frac{115 + j15}{50}$$

$$Z_L = 2.09 + j1.36$$

$$\lambda = 1.183 \text{ cm}$$

$$\text{stub location} = 0.205 \lambda$$

$$= 0.205 \times 1.183 = 0.242 \text{ cm}$$

$$\text{stub length} = 0.11 \lambda$$

$$= 0.11 \times 1.183 = 0.130 \text{ cm}$$

5. on a transmission line a standing wave pattern is observed and VSWR is found to be 3 and it is also noted that two successive voltage minima are 30cm apart. The first voltage minimum is found to be 12cm from the load. The length of the line is 85.2cm and Z_0 is 200Ω . Find

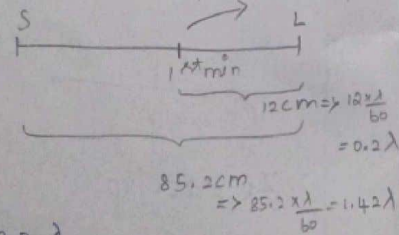
i) sending end impedance ii) load impedance.

soln: Given,

$$S = 3$$

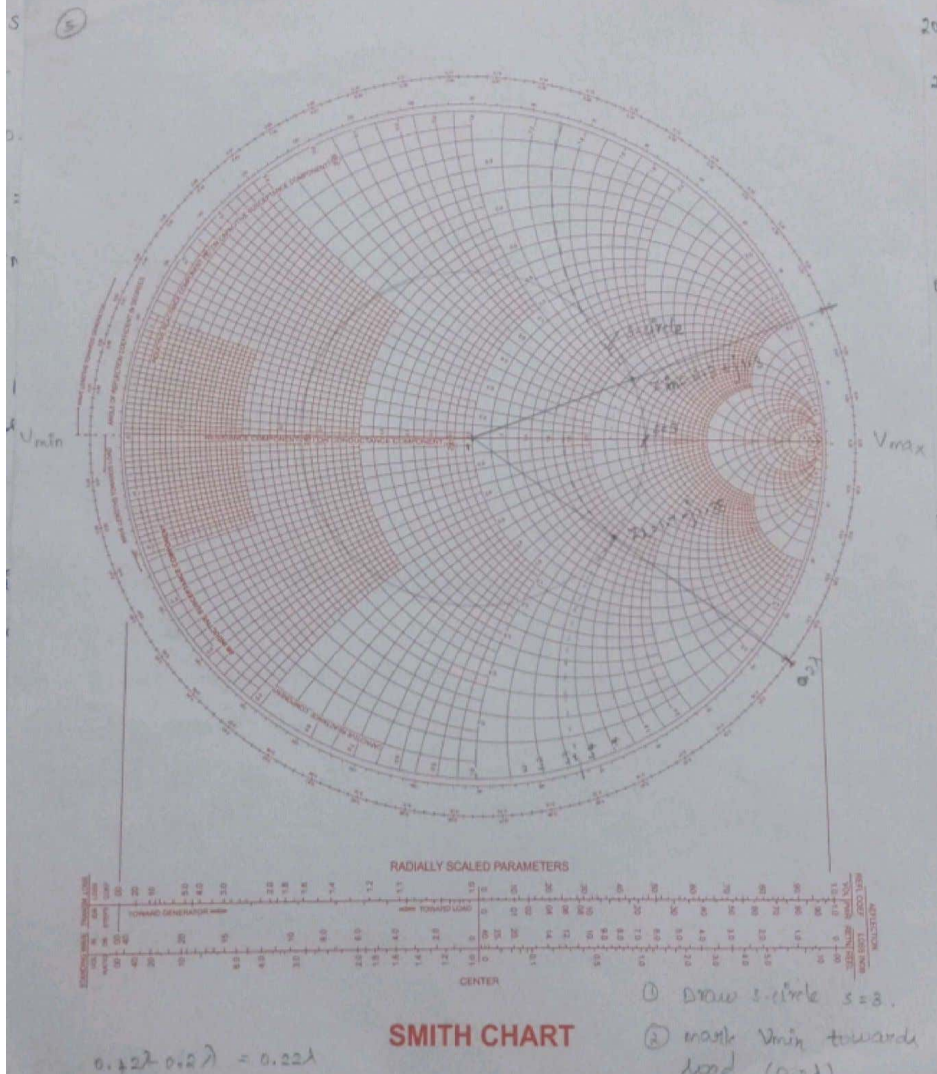
$$\frac{\lambda}{2} = 30\text{cm}$$

$$\lambda = 60\text{cm} \Rightarrow 1\text{cm} = \frac{\lambda}{60}$$



i) $I_{IN} = I_{in} Z_0 = (2.3 + j1.3) 200 = 460 + j260 \Omega$

ii) $I_L = I_{\lambda} Z_0 = (1.7 - j1.35) 200 = 340 - j270 \Omega$



b A 75Ω lossless transmission line is to be matched with a $100 - j80 \Omega$ load using single stub. Calculate stub length and its distance from the load correspond to the frequency of 30 MHz using Smith chart.

Sol:

$$Z_0 = 75 \Omega$$

$$Z_L = 100 - j80 \Omega$$

$$f = 30 \text{ MHz} = 30 \times 10^6$$

$$Z_L = Z_0 \Gamma_L$$

$$\Gamma_L = \frac{Z_L}{Z_0}$$

$$= \frac{100 - j80}{75}$$

Normalized load impedance,

$$\Gamma_L = 1.33 - 1.06j$$

$$\lambda = \frac{c}{f} = \frac{3 \times 10^8}{30 \times 10^6}$$

$$= 10 \times 10^2 = 10 \text{ m}$$

$$= 1000 \text{ cm}$$

$$Y_L = \frac{1}{1.33 - 1.06j} = 0.257 + 0.365j$$

$$b = 0.096\lambda$$

$$= 0.096 \times 10$$

$$b = 0.96 \text{ m}$$

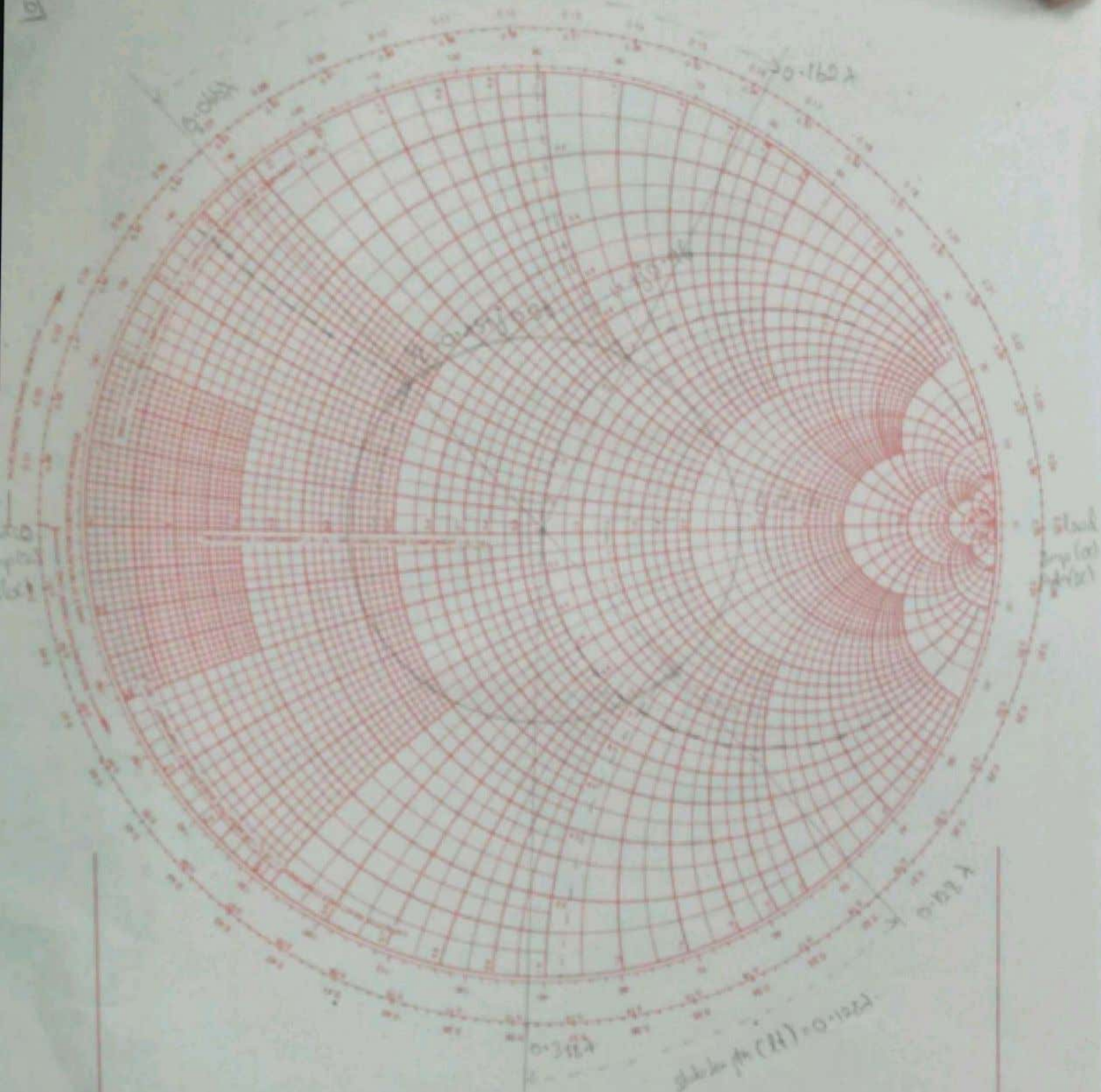
$$L_t = 0.128\lambda$$

$$= 0.128 \times 10$$

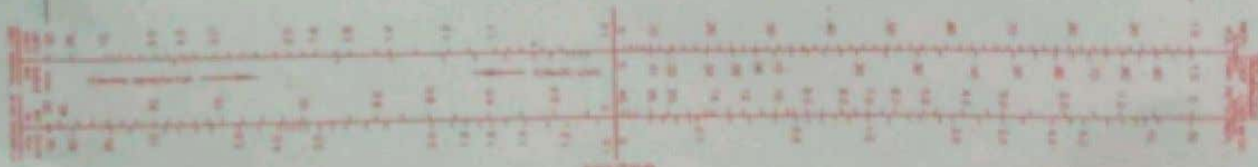
$$L_t = 1.28 \text{ m}$$

$$d_{\text{delay}}(\lambda) = 0.096\lambda$$

$$0.165\lambda$$



RADIALLY SCALED PARAMETERS



SMITH CHART